

NetraSetu — ML Benchmarks

Prepared for SIH portal submission. Every number below is measured on real data by real code already in this repository — nothing is estimated or projected. Each figure cites the exact file it came from so it can be independently reproduced. Where a number is optimistic for a stated reason (calibration contamination, small sample size, etc.), that caveat is stated next to the number, not omitted — this project's own standing rule is that a reported result that hasn't been cleanly validated says so, rather than reading as more solid than it is.

Served model versions (per docs/RELEASE.md , confirmed live): classifier branchA_v2c , vessel segmentation vessel_unet_v1 , optic-disc/fovea localization localization_v1 , hard-exudate segmentation bright_lesion_unet_v1 , red-lesion (microaneurysm+haemorrhage) segmentation red_lesion_unet_v2 (confirmed live — see §5).

1. Headline result — DR severity classification (Branch A, deployed model)

Independent run of the served classifier (branchA_v2c , calibrated, the exact branchAInfer.py code path used in production) against the **103 official IDRiD test-set images** and their ground-truth grades. Reproducible with python scripts/audit/eval_idrid_test.py .

Metric	Result	SIH problem-statement target
Referable-DR (grade ≥ 2) sensitivity — live referral flag	100% (64/64)	> 90% 
Referable-DR sensitivity — by raw grade argmax	96.9%	
Referable-DR specificity — live referral flag	82.1%	> 85%  (below target)
Referable-DR specificity — by raw grade argmax	79.5%	
Quadratic-weighted kappa (QWK)	0.841	
Exact-grade accuracy	77.7%	
Within-one-grade accuracy	91.3%	
Grade-4 (proliferative DR) recall	13/13 (100%)	
Confidence-tier distribution (of 103)	Tier A: 4 · Tier B: 70 · Tier C: 29	
Tier-A cases that were actually referable (false auto-clear)	0	
Conformal prediction sets that are contiguous (no gaps, e.g. not {0,3})	83.5%	

Reference point, not a like-for-like comparison: the FDA-pivotal-trial benchmark for autonomous DR screening (IDx-DR, 900 patients, 10 primary-care sites) reported 87.2% sensitivity / 90.7% specificity for referable DR. NetraSetu is a decision-support system with a human ophthalmologist confirming every result, not an autonomous diagnostic, so this is a scale reference, not a claim of equivalence.

Read this as a smoke test on the deployed model, not a clean validation. The temperature scaling and referral threshold were fit on a pool that includes these same 103 test images (calibration_branchA_v2c.json : "pooled val+test, n=1161") — source: docs/STALE_CLAIMS_AUDIT.md §3.1. What this table *does* establish: the deployed model is not broken, no case in this set was silently under-graded to a false negative, and grade-4 recall — previously a known ship-blocking weakness at 0.444 on an older

model version — is resolved on this population. What it does *not* establish: a clean, held-out, publication-grade accuracy number, and specificity is honestly below the target even though sensitivity clears it — meaning only about 4% of cases fully auto-clear, and roughly 96% still reach a human reviewer at this calibration.

Prior model versions, for context (`central-system/backend/ml-pipeline/diagnostics/MODEL_INTERFACE_REFERENCE.md`): v1 (384×384 input): test QWK 0.869, referable sensitivity/specificity 0.86/0.94, grade-4 recall 0.444 — this is the version v2c replaced specifically because of that grade-4 weakness.

2. Statistical safety validation — conformal confidence tiers

Class-conditional conformal calibration determines which cases can safely skip human review (Tier A). This is evaluated separately from raw accuracy because it's a *coverage guarantee*, not a point estimate: source `central-system/backend/ml-pipeline/diagnostics/out/conformal_v3_crossfit_report_branchA_v2c.json` , a cross-fit study on the pooled val+test population (n=1161).

Metric	Result	Gate
Referable-case coverage (lower confidence bound)	0.9416	≥ 0.93 ✓
False auto-clear rate, referable cases	0.0000	≤ 0.05 ✓
False auto-clear rate, grade ≥ 3 cases	0.0000	≤ 0.02 ✓
Mean prediction-set size	2.01	(smaller = more decisive)
Tier distribution (pooled)	A: 38.4% · B: 43.8% · C: 17.8%	
Grade-4 cases auto-cleared to Tier A, across 1000 fold-assignments	0	
Pooled sensitivity (95% CI)	0.9503 [0.9277, 0.9661] (478/503)	
Pooled specificity (95% CI)	0.9103 [0.8861, 0.9298] (599/658)	

The clinical safety claim this supports: across a cross-fit study covering the model's full calibration population, **no grade-4 case has ever auto-cleared without human review**. This is the property the tiering system exists to guarantee, and it held with zero exceptions.

Cross-implementation parity: the calibrated classifier's Python and MATLAB inference paths were checked against each other on 10 real IDRiD images — exact grade and tier agreement 10/10, max calibrated-probability difference 5.14×10^{-7} (effectively floating-point noise, not a real divergence). Source: same report file.

3. Segmentation models

Model	Role	Test set	n	Dice	Sensitivity	Specificity	IoU
Vessel U-Net (vessel_unet_v1)	Vessel map — feeds NV suspicion, MA false-positive suppression	CHASE_DB1 (training domain)	28	0.8024	0.799	0.986	0.670
Vessel U-Net (same model, no retraining)	Cross-dataset generalization check	DRIVE (unseen domain)	20	0.6186	0.463	0.995	0.448
Hard-exudate U-Net (bright_lesion_unet_v1)	Bright lesion segmentation	IDRiD heldout	27	0.6676	0.723	0.994	0.501
Red-lesion U-Net, v1 (single-class MA+HE)	Predecessor to the deployed v2 model	IDRiD heldout	16	0.6105	0.570	0.997	0.439

Source: central-system/backend/ml-pipeline/diagnostics/out/m2_metrics.json , m45_metrics.json .

Optic disc / fovea localization (localization_v1), IDRiD heldout, n=77 (diagnostics/out/m3_metrics.json):

Target	Mean error (native px)	Mean error (512-space px)	Within one disc radius
Optic disc	26.9	4.07	98.7%
Fovea	85.6	12.2	96.1%

A dedicated peak-confidence gate (threshold 0.37, chosen as "the smallest threshold with 100% validation gross-miss sensitivity" — diagnostics/out/fovea_gate_report.json) catches the cases where fovea localization has no real confidence peak, marks them fovea_unreliable , and routes them to mandatory review instead of grading on a silently-wrong quadrant axis.

Important honesty note on the red-lesion model — read before quoting a number: The model actually running in production for microaneurysm/haemorrhage detection is **v2** (a 3-class architecture separating microaneurysms from haemorrhages, each with its own minimum-lesion-size filter) — confirmed live: segInfer.py:116 defaults RED_LESION_MODEL_VERSION to "v2" , no environment override exists, and real production database rows carry non-null, non-zero microaneurysm/haemorrhage counts under redLesionModelVersion: "v2" . As of 2026-09-30 it also runs through the **MATLAB session** by default, like every other segmentation model (the ONNX-imported red_lesion_unet_v2.mat , parity-checked against its PyTorch source in training/parityCheckRedLesionV2.m), with the original PyTorch checkpoint kept only as an explicit fallback path. **However, v2 has no independently measured Dice/sensitivity score of its own of its own anywhere in this codebase** — the 0.6105 Dice figure in the table above is v1's, a different (single-class) architecture it replaced. v2's improvement over v1 is architectural and qualitative (it can report a microaneurysm count and a haemorrhage count separately, which v1 could not do at all — v1 could only report a combined red-lesion total), not yet backed by its own quantitative accuracy benchmark. **Do not state a Dice/sensitivity number for v2 specifically — none exists yet.** This is worth commissioning before the next round, not before this submission.

Not detected, by explicit design decision, not a bug: cotton-wool spots (soft exudates) have no dedicated detector this round — the available pixel-level training data for this lesion type was judged too sparse to support a real detector, so this field is always reported as unmeasured, never as a false zero.

Neovascularization is reported as a suspicion score (vessel-density/branching/tortuosity proxy near the optic disc), not a segmentation claim, because it has the least public pixel-level training data of any lesion type here. Its own measured performance (AUC 0.286 on IDRiD, 0.379 on Messidor-2 — both **below chance**, source docs/STALE_CLAIMS_AUDIT.md §3.4) means it is

currently gated off and does not influence any grade or tier — a CNN-predicted grade-4 triggers mandatory review on its own, independent of what this score says.

4. Rule engine (Branch B — explicit ICDR/ETDRS criteria)

Branch B is plain, testable code implementing the standard ICDR "4-2-1" rule directly on quadrant-mapped lesion counts (no learned weights), so it can be audited against the clinical rule text line by line rather than trusted as a black box. Live thresholds (frozen, not to be changed without an explicit decision — `CLAUDE.md`): `RED_FLOOR=3` , `GRADE3_QUAD_MIN=3` , `RULE_MAX_GRADE=3` .

No system-wide Branch A/B agreement-rate statistic is reported here — the only file that looks like one (`diagnostics/out/agreement_results.csv`) is a small, curated set of disagreement examples used for debugging, not a representative sample, and computing a percentage from it would misrepresent what it measures. When the two branches disagree, the case is forced into mandatory review with a required explicit grade resolution — a disagreement is treated as a safety signal, not noise to be averaged away, so this system doesn't need a single agreement percentage to make its safety case.

A separate, **proposed but not adopted** recalibration study exists (`diagnostics/out/gate2_recalibration_report.json`) exploring alternate thresholds on IDRiD; it is explicitly marked "not read by any production code yet" and is not what ships.

5. System-level reliability (not an accuracy metric)

A full-dataset backend soak test (`scripts/test-full-dataset.js`) submitted every IDRiD image present on disk (447 images) through the real capture → quality-gate → sync → grading pipeline end-to-end, exactly as a PHC technician's submission would flow:

Metric	Result
Images submitted	447
Successfully graded	447 (100%)
Failed	0
Timed out	0

This is a throughput/reliability result (the pipeline completes without crashing or dropping a case under real load), not a claim about grading accuracy — accuracy is reported separately in §1.

6. Honest limitations (stated on purpose, not omitted)

- Specificity (82.1%) is below the >85% target even though sensitivity clears it — only ~4% of cases fully auto-clear at the current calibration.
- The §1 and §2 numbers are measured on a population that overlaps with the model's own calibration set — a genuinely clean, held-out validation has not been run.
- Messidor-2 (external, different-population validation) was used for model *selection* between v2a/v2b/v2c, not as an untouched external test set — so it does not substitute for one.
- The red-lesion model actually running in production (v2) has no independently measured accuracy score of its own — see §3.

- Neovascularization detection is measured and has failed (below-chance AUC) — it is gated off, not merely unmeasured.
- Cotton-wool spot (soft exudate) detection is out of scope this round, not a weak metric.
- Vessel segmentation shows a real cross-domain gap (Dice 0.80 in-domain vs. 0.62 on an unseen camera/dataset) — a live, fixed 0.5 threshold is used, not a per-domain-adjusted one.
- No system-wide Branch A/B agreement rate has been measured on a representative sample.